UNIT 1 · 9 hours

Analysis of RL and RC Series Circuits



Circuit Fundamentals

Basic Electrical and Electronics Engineering (EEE103)
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NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

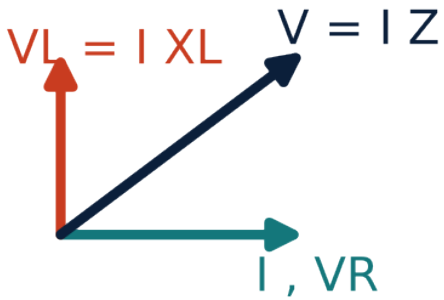
- Draw phasor and impedance diagrams for RL and RC series circuits
- Compute impedance, current, phase angle and power
- Decide whether current lags or leads
- Solve numericals

Syllabus context

Analysis of R, L, C, RL, RC and RLC series circuits, real power, reactive power, apparent power and power factor.

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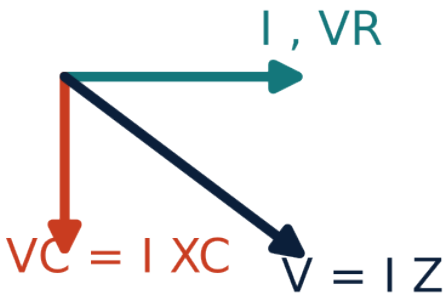
RL series circuit: impedance and phase



- Same current flows in R and L; use I as reference.
- VR = IR in phase with I; VL = IXL leads by 90°.
- $V = \sqrt{(VR^2 + VL^2)} = I\,Z$
- $Z = \sqrt{(R^2 + XL^2)}$, $\varphi = \tan^{-1}(XL/R)$
- Current lags the supply voltage by φ.

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RC series circuit: impedance and phase

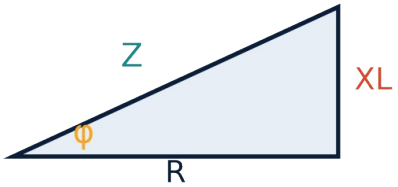


- VR = IR in phase with I; VC = IXC lags by 90°.
- $V = \sqrt{(VR^2 + VC^2)} = I\,Z$
- $Z = \sqrt{(R^2 + XC^2)}$, $\varphi = \tan^{-1}(XC/R)$
- Current leads the supply voltage by φ.

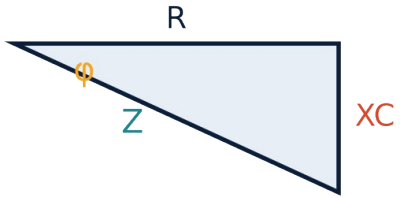
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Impedance triangles

R–L series: current lags



R–C series: current leads



- Z is the phasor sum of R and X.
- cos φ = R/Z = power factor.
- tan φ = X/R.

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Power in RL and RC circuits

$$P = V I \cos \varphi$$

active power (W)

$$P = I^2 R$$

only R dissipates

- Instantaneous power $p = \tfrac{1}{2} V_m I_m \cos \varphi - \tfrac{1}{2} V_m I_m \cos(2\omega t - \varphi)$.
- The second term averages to zero, so the average is $VI \cos \varphi$.
- Positive power lobes are larger than negative ones: net power is absorbed.

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Numerical: RL series

Problem R = 30 Ω and L = 0.1 H in series across 230 V, 50 Hz. Find Z, I, φ, power factor and P.

- $X_L = 2\pi \cdot 50 \cdot 0.1 = 31.42\ \Omega$
- $Z = \sqrt{(30^2 + 31.42^2)} = 43.44\ \Omega$
- $I = 230/43.44 = 5.29$ A; $\varphi = \tan^{-1}(31.42/30) = 46.3°$ (lagging)
- $\cos\varphi = 30/43.44 = 0.69$; $P = I^2R = 5.29^2 \times 30$

Answer: I = 5.29 A, pf = 0.69 lag, P = 840 W

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Numerical: RC series

Problem R = 50 Ω and C = 50 µF in series across 230 V, 50 Hz. Find Z, I and φ.

- $X_C = 1/(2\pi \cdot 50 \cdot 50 \times 10^{-6}) = 63.66\ \Omega$
- $Z = \sqrt{(50^2 + 63.66^2)} = 80.95\ \Omega$
- $I = 230/80.95 = 2.84\ A$
- $\varphi = \tan^{-1}(63.66/50) = 51.9°$ (current leads)

Answer: Z = 80.95 Ω, I = 2.84 A, φ = 51.9° leading

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Key takeaways

- $Z = \sqrt{(R^2 + X^2)}$, $V = IZ$
- RL: current lags; RC: current leads
- pf = cos φ = R/Z
- Only the resistance dissipates power: $P = I^2R$

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Practice questions

Q1. R = 10 Ω, L = 50 mH in series on 100 V, 50 Hz. Find I, φ, P.

Q2. R = 20 Ω, C = 100 µF on 230 V, 50 Hz. Find Z, I, pf.

Q3. A coil has R = 8 Ω, XL = 6 Ω. Find Z and pf.

Q4. Draw the phasor diagram for an RC circuit with I as reference.